

Einstein's General Theory of Relativity

Lecture 3: Gravitational Redshift, Tensors, and the Metric

Lectures by Leonard Susskind

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Chapter 1

Gravitational Redshift, Tensors, and the Metric

1.1 What Gravity Does to a Vertical Light Ray

We return to the elevator and turn the light ray through ninety degrees. A bullet fired upward loses kinetic energy as it rises and therefore slows down. Light cannot behave in quite that way. Every local observer measures its speed to be c , so the quantity that changes as the ray climbs is its frequency. Upward propagation produces a redshift; downward propagation produces a blueshift.

The accelerating elevator gives the quickest derivation. Let the elevator be momentarily at rest when a photon is emitted from its roof toward the floor. During the light-travel time the elevator accelerates upward. The floor therefore moves toward the arriving photon, and the detector sees a Doppler blueshift.

If the elevator height is h , then to first order

$$\Delta t \simeq \frac{h}{c}, \quad \Delta v \simeq g\Delta t \simeq \frac{gh}{c}. \quad (1.1)$$

The first-order Doppler shift is consequently

$$\frac{\Delta\nu}{\nu} \simeq \frac{\Delta v}{c} \simeq \frac{gh}{c^2}. \quad (1.2)$$

For a downward ray this quantity is positive. Reversing the experiment reverses the sign:

$$\frac{\nu_{\text{floor}}}{\nu_{\text{roof}}} \simeq 1 + \frac{gh}{c^2}, \quad \frac{\nu_{\text{roof}}}{\nu_{\text{floor}}} \simeq 1 - \frac{gh}{c^2}. \quad (1.3)$$

Question from the audience. Why do the Doppler shifts not cancel as they would for source and receiver moving together? ^a

Response. A constant-velocity elevator has one inertial rest frame, and source and receiver remain at rest in it. Here emission and reception occur at different times while the elevator is accelerating. The source is momentarily at rest at emission, but the receiver has acquired velocity by reception. That change of velocity is precisely what survives in Eq. (1.2).

^aLecture timestamp: 00:02:23–00:04:55.

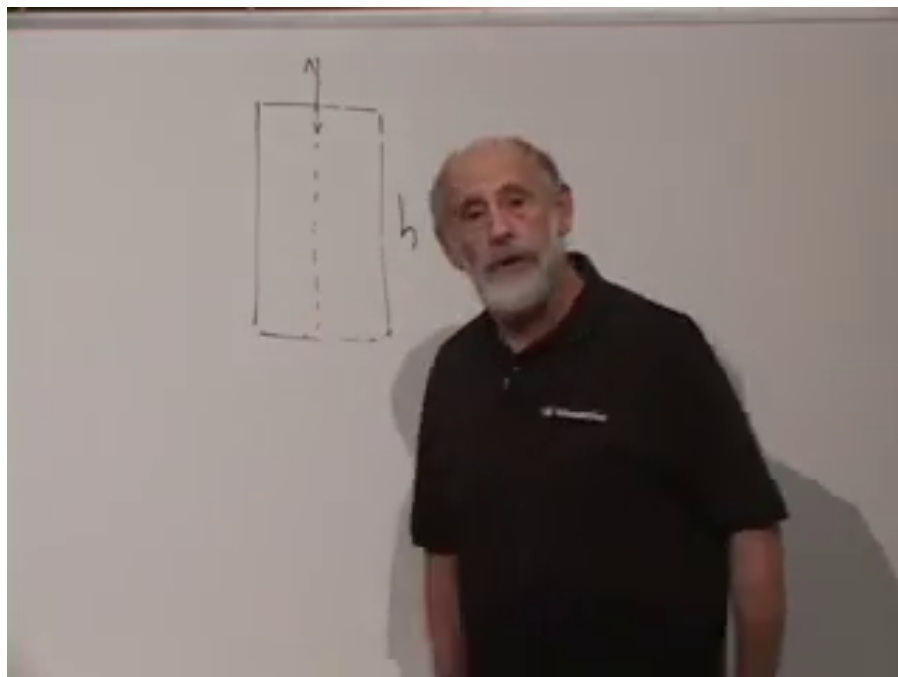


Figure 1.1: The vertical-light experiment in an elevator of height h . During the photon's flight, upward acceleration changes the velocity of the receiving end of the elevator.

Lecture frame: 00:05:00.

The equivalence principle now turns the kinematic result into a gravitational one. A laboratory supported in a uniform gravitational field is locally equivalent to the upward-accelerating elevator. Light sent downward is therefore blueshifted, while light sent upward is redshifted. The effect is small near Earth, but it is measurable in a sufficiently sensitive tower experiment.

The same reasoning applies to light escaping from the Sun. Its spectral lines arrive redshifted relative to identical lines measured in the laboratory. For a spherical source of mass M and radius R , the weak field scale is

$$z \equiv \frac{\nu_{\text{emit}} - \nu_{\text{receive}}}{\nu_{\text{receive}}} \simeq \frac{GM}{Rc^2} = \frac{v_{\text{esc}}^2}{2c^2}. \quad (1.4)$$

Large radius alone is not decisive. What matters is compactness, or equivalently the escape speed. White dwarfs and neutron stars produce larger shifts than ordinary stars, and the redshift becomes extreme near a black-hole horizon.

Question from the audience. Would a physically larger star necessarily produce a larger redshift? ^a

Response. Not necessarily. The controlling dimensionless combination is $GM/(Rc^2)$, not M or R separately. A compact object can have a much larger surface redshift than a more massive but diffuse object.

^aLecture timestamp: 00:06:00–00:07:35.

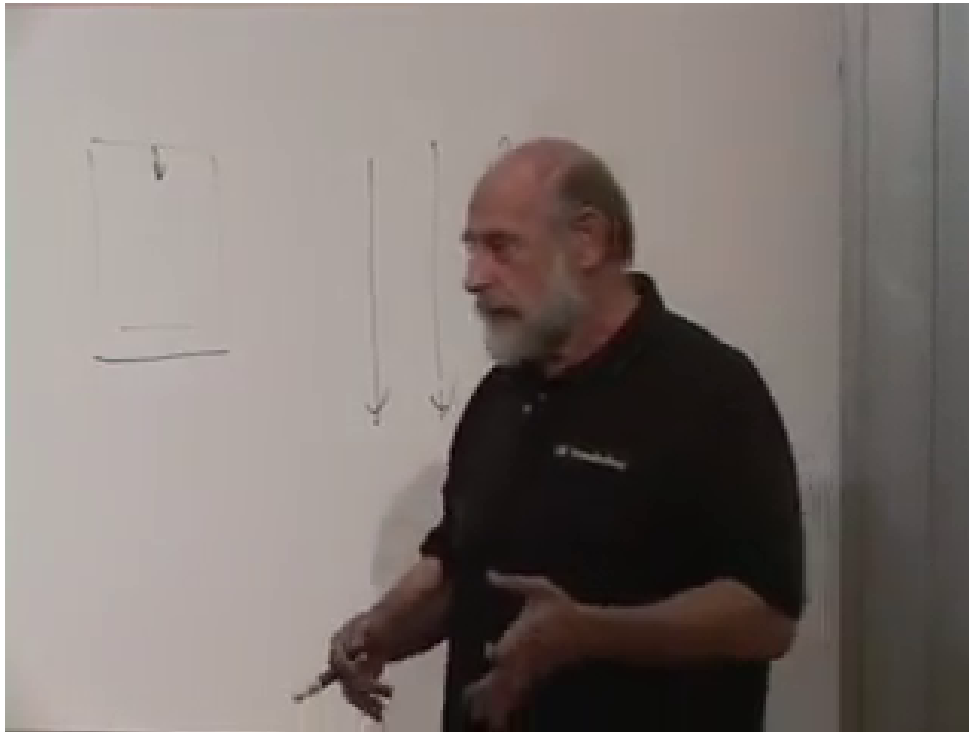


Figure 1.2: The freely falling elevator. Gravitational blueshift of the descending light and the receiver's kinematic redshift cancel in the local experiment.

Lecture frame: 00:12:05.

Question from the audience. Does the frequency shift contradict the equivalence principle by revealing gravity inside the elevator? ^a

Response. No. The supported elevator is equivalent to an accelerated laboratory, not to an inertial one. The shift is the same observable described either as a Doppler effect in the accelerating frame or as a gravitational frequency shift in the supported laboratory.

^aLecture timestamp: 00:08:30–00:09:22.

1.2 Free Fall and the Local Disappearance of Gravity

Now release the elevator. Send a photon downward while the entire laboratory falls. Relative to observers held fixed in the gravitational field, the photon gains frequency as it descends. At the same time the floor falls away from the photon and produces an opposing kinematic redshift. In a sufficiently small freely falling laboratory the two effects cancel:

$$\left(\frac{\Delta\nu}{\nu}\right)_{\text{grav}} + \left(\frac{\Delta\nu}{\nu}\right)_{\text{Doppler}} = 0. \quad (1.5)$$

An exact calculation must use special-relativistic Doppler factors. A Newtonian expansion establishes the leading cancellation but can leave spurious higher-order terms. The exact local cancellation is what the equivalence principle requires.

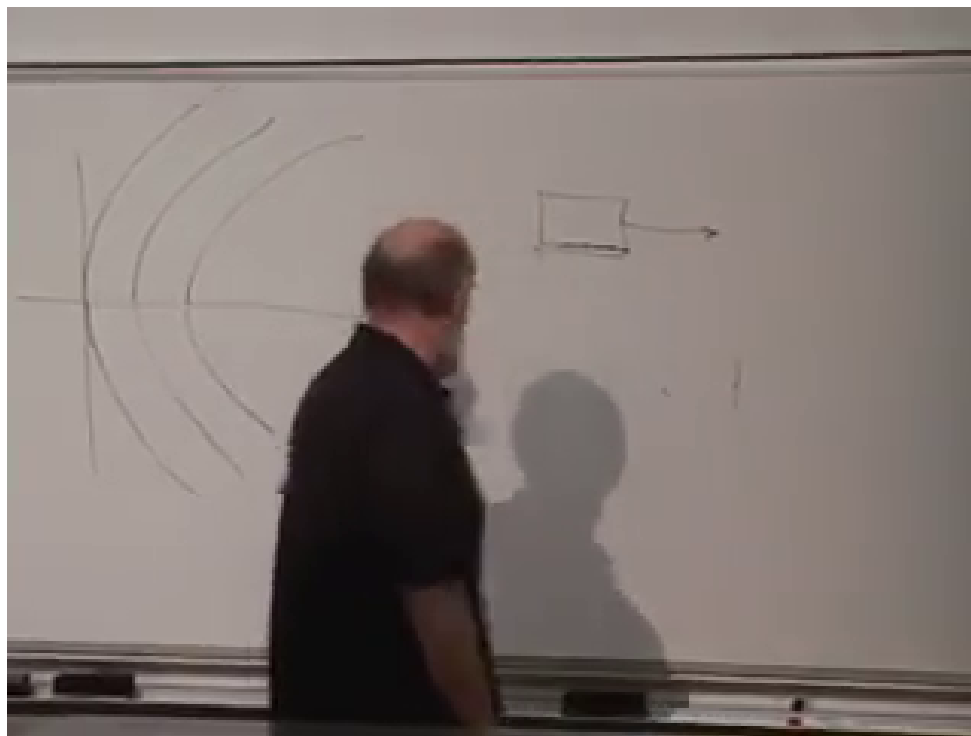


Figure 1.3: Curved trajectories traced by the meter sticks of an accelerating coordinate system. Apparent gravity can arise from the coordinates even when the underlying geometry is flat.

Lecture frame: 00:14:25.

Question from the audience. Why should a falling detector see no net shift when both the photon and the elevator are accelerating? ^a

Response. The two accelerations enter differently. Relative to a supported frame, gravity blueshifts the downward photon, while the falling floor moves away from it and redshifts the received signal. In the local freely falling frame these are not two independent effects; they are the two descriptions whose cancellation makes that frame locally inertial. Only variations of gravity across the apparatus can remain.

^aLecture timestamp: 00:09:22–00:12:48.

This qualification, *locally*, is essential. A uniform acceleration can be absorbed into the coordinates used to describe a small laboratory. In an inertial picture, the meter sticks fixed in an accelerating elevator trace a family of curved trajectories. The resulting coordinate grid is curved even though the underlying spacetime used in this preliminary argument is flat.

The field of a real gravitating body is not uniform over an extended region. Its direction changes from place to place, and its strength changes with distance. One freely falling elevator can match the field near one chosen trajectory, but no single elevator removes the field everywhere. Neighboring freely falling bodies acquire relative acceleration: the tidal effect.

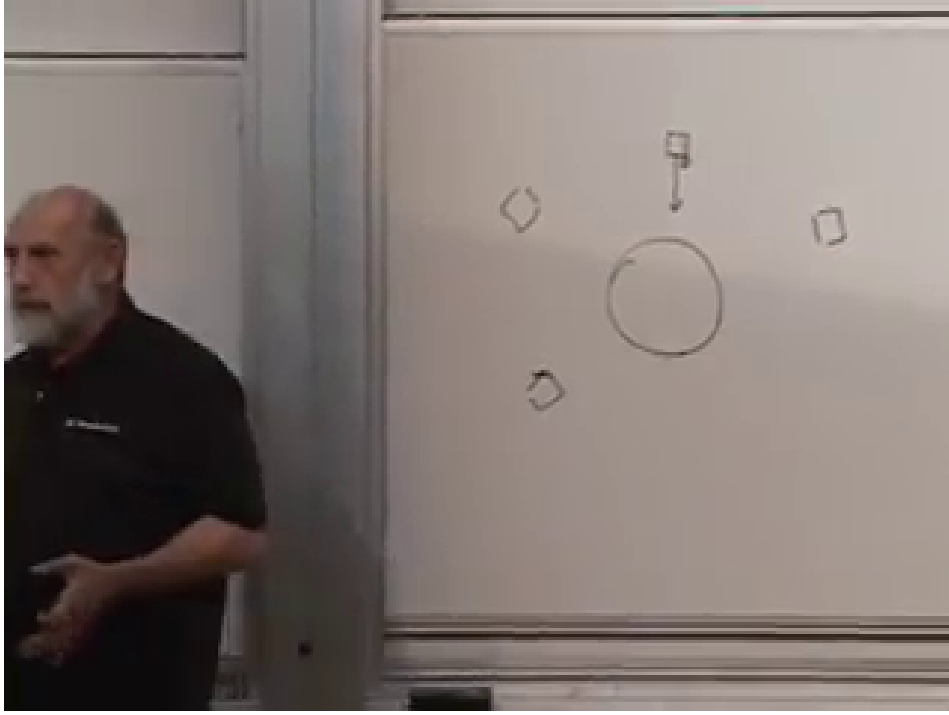


Figure 1.4: Several local laboratories around a gravitating body. Their acceleration directions differ, displaying why one global accelerated frame cannot remove the whole field.

Lecture frame: 00:17:40.

Question from the audience. Is this already the Rindler coordinate construction? ^a

Response. It is the same physical direction, but the present discussion is deliberately pre-relativistic. The exact Rindler chart requires special relativity and will be treated only after the geometric language is in place.

^aLecture timestamp: 00:18:00–00:18:55.

We have therefore reached the first geometric statement of general relativity. Gravity that is common to an infinitesimal freely falling laboratory can be transformed away. Tidal gravity cannot. The obstruction to removing the latter will become the curvature of spacetime. To describe that obstruction without tying ourselves to one coordinate grid, we need tensor calculus.

1.3 Coordinates, Scalars, and Infinitesimal Changes

Begin with ordinary d -dimensional space, before adding time. Label a point by

$$x = (x^1, x^2, \dots, x^d), \quad (1.6)$$

and let $\phi_{(x)}(x)$ be a scalar field. An infinitesimal displacement has coordinate components $dx^1, \dots, dx^d$. The corresponding change of the scalar is its total differential,

$$d\phi = \sum_{m=1}^d \frac{\partial \phi}{\partial x^m} dx^m. \quad (1.7)$$

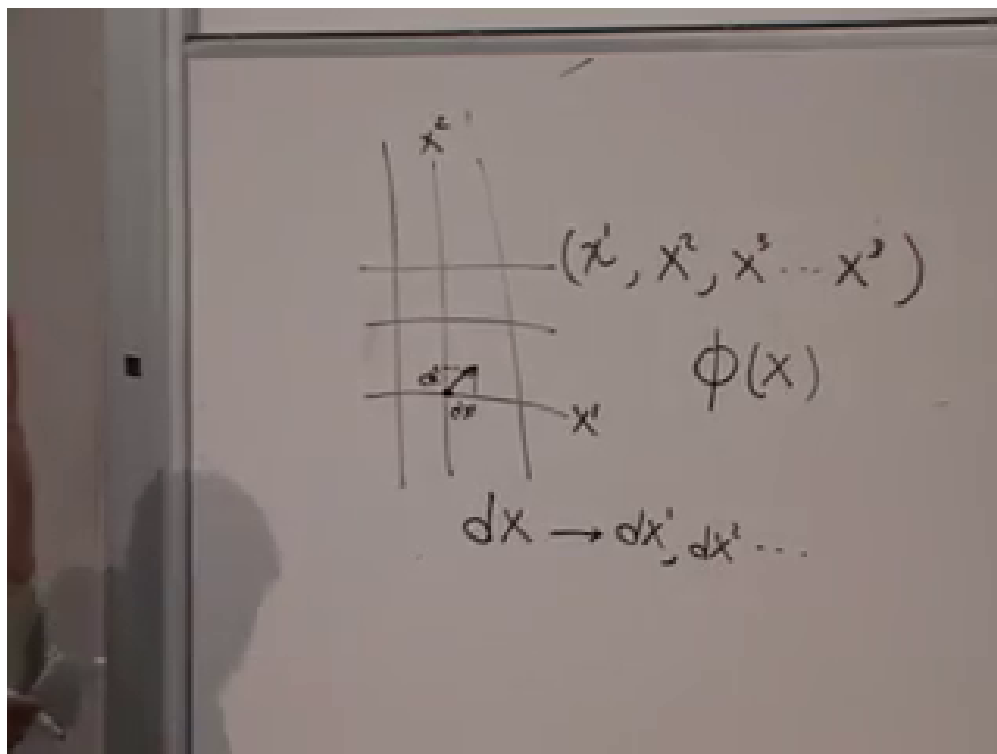


Figure 1.5: A d -dimensional coordinate system, a scalar field $\phi(x)$, and the infinitesimal displacement dx used in the total differential.

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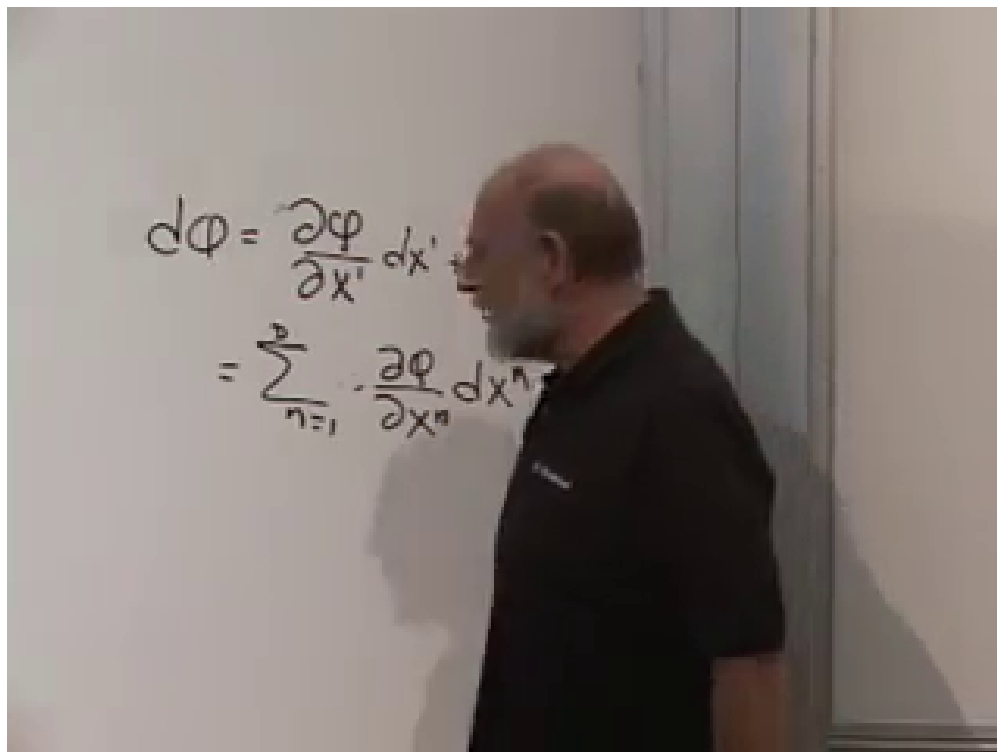


Figure 1.6: The total differential written first as an explicit sum and then in the repeated-index convention.

Lecture frame: 00:25:40.

From now on a repeated index is summed unless stated otherwise. With this Einstein convention,

$$d\phi = \frac{\partial \phi}{\partial x^m} dx^m. \quad (1.8)$$

The repeated m is a dummy index. Renaming it r changes nothing, provided the same label is used in both factors and does not collide with a free index.

Question from the audience. Does the order of the differentials matter, and is $d\phi$ a finite change? ^a

Response. The components here are ordinary commuting numbers, so their products commute. The symbol d denotes the first-order differential. For a finite displacement Δx , higher-order terms generally enter; Eq. (1.8) is the linear infinitesimal relation.

^aLecture timestamp: 00:27:30–00:30:08.

Introduce another coordinate system,

$$y^n = y^n(x^1, \dots, x^d). \quad (1.9)$$

An admissible change of coordinates is one-to-one on the region being used, differentiable, and equipped with a differentiable inverse. It relabels points; it does not change the dimension of the space. The chain rule gives

$$\frac{\partial \phi}{\partial y^n} = \frac{\partial \phi}{\partial x^m} \frac{\partial x^m}{\partial y^n}. \quad (1.10)$$

$$\frac{\partial \varphi}{\partial y^n} = \frac{\partial \varphi}{\partial x^m} \frac{\partial x^m}{\partial y^n} = \frac{\partial \varphi}{\partial x^1} \frac{\partial x^1}{\partial y^n} + \frac{\partial \varphi}{\partial x^2} \frac{\partial x^2}{\partial y^n} + \dots$$

Figure 1.7: The scalar chain rule between the x and y coordinate systems. The repeated index m is summed while n remains free.

Lecture frame: 00:35:05.

For each free index n , the repeated index m is summed.

The direct and inverse Jacobians satisfy

$$\frac{\partial x^m}{\partial y^n} \frac{\partial y^n}{\partial x^r} = \delta^m_r. \quad (1.11)$$

This identity is the coordinate version of multiplying a nonsingular matrix by its inverse.

Question from the audience. What restrictions are imposed on a coordinate transformation? ^a

Response. Within the region under discussion, each point must have a unique label and the map must be smooth enough to differentiate. Its Jacobian must be nonsingular so that the inverse coordinates exist. A chart may fail at a boundary or coordinate singularity, but it must be regular where the tensor formulas are used.

^aLecture timestamp: 00:30:08–00:31:48.

Question from the audience. Can a coordinate change alter the number of coordinates? ^a

Response. No. Both charts describe the same d -dimensional space. They have the same number of independent coordinates, and the inverse transformation returns one set from the other.

^aLecture timestamp: 00:37:45–00:38:55.

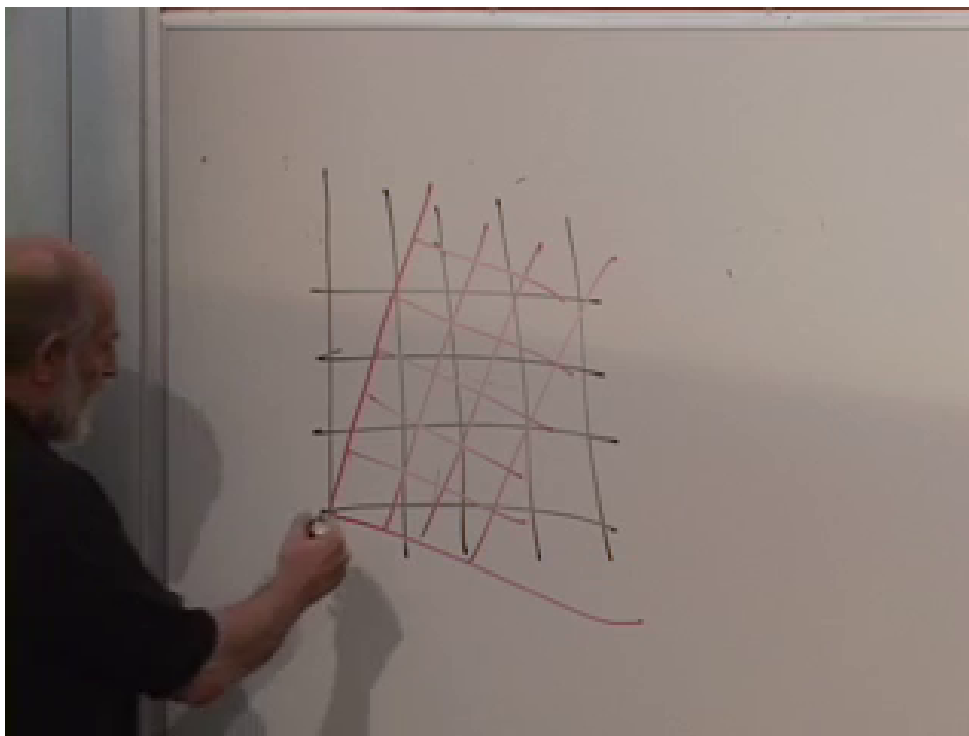


Figure 1.8: Two coordinate grids on the same plane. The red skew grid changes component values without changing the geometric objects being described.

Lecture frame: 00:40:10.

1.4 Geometric Objects and Their Components

Draw two grids on the same plane, one rectangular and one skew. The points, curves, and arrows in the plane have not changed, but their numerical coordinates have. Tensor notation separates the geometric object from the components used to represent it.

A scalar is the simplest example. It assigns one coordinate-independent number to each point. If x and y label the same point, the two coordinate representations obey

$$\phi_{(y)}(y) = \phi_{(x)}(x(y)). \quad (1.12)$$

This is what it means to call a scalar a rank-zero tensor.

The infinitesimal displacement is the model contravariant vector. Applying the total-differential rule to each coordinate function $y^n(x)$ gives

$$dy^n = \frac{\partial y^n}{\partial x^m} dx^m. \quad (1.13)$$

Any contravariant vector transforms by the same rule:

$$\boxed{V_{(y)}^n = \frac{\partial y^n}{\partial x^m} V_{(x)}^m.} \quad (1.14)$$

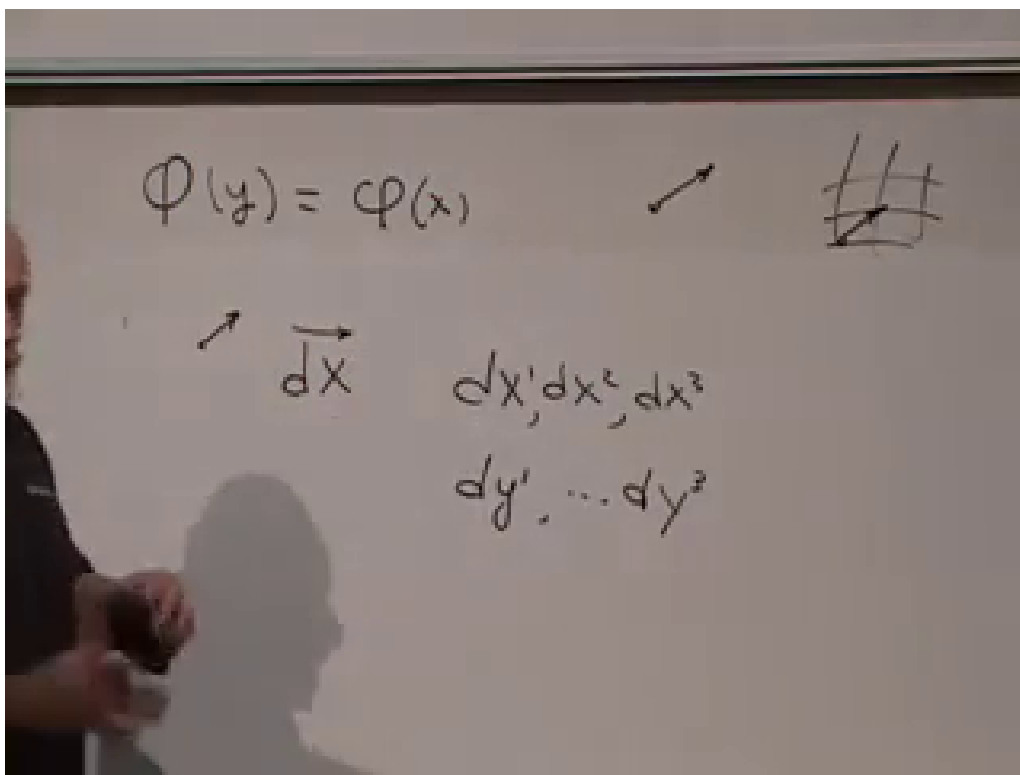
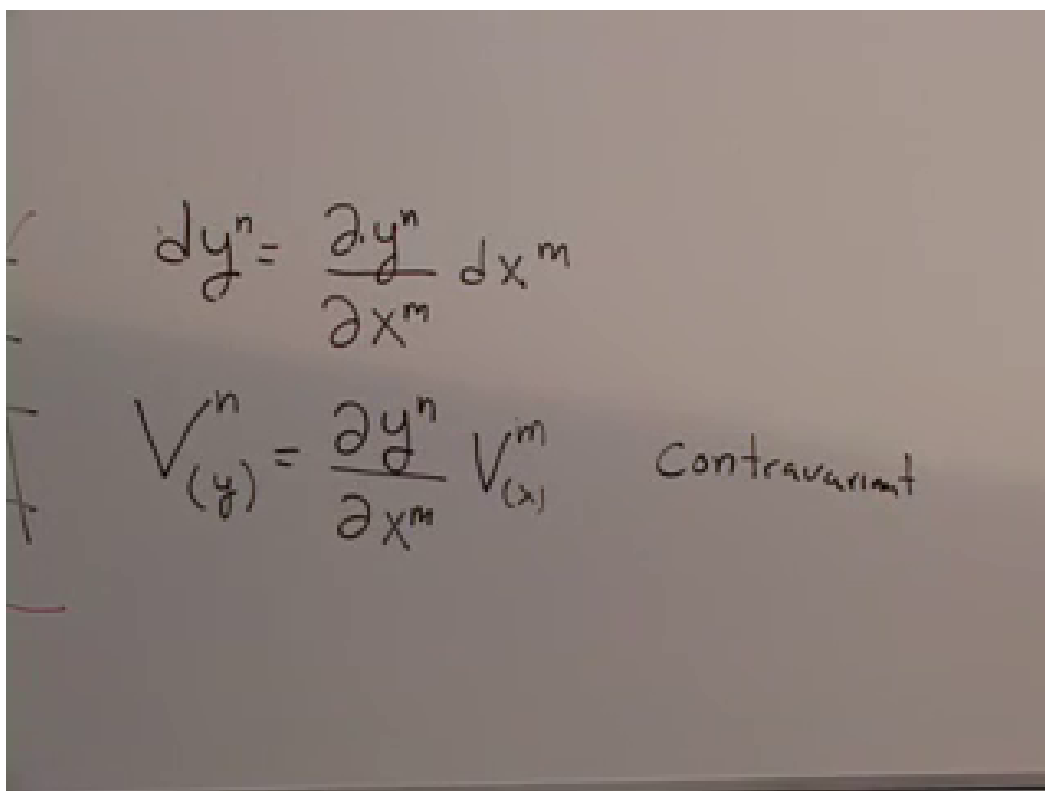


Figure 1.9: Scalar invariance and a free displacement vector. The scalar has the same value at the same point, while vector components change with the coordinate grid.

Lecture frame: 00:47:30.



The image shows a chalkboard with two handwritten equations. The first equation is the displacement law: $dy^n = \frac{\partial y^n}{\partial x^m} dx^m$. The second equation is the transformation of a contravariant vector: $V_{(y)}^n = \frac{\partial y^n}{\partial x^m} V_{(x)}^m$, followed by the word "Contravariant" written to the right of the equation.

Figure 1.10: The displacement law and the general transformation of a contravariant vector. The Jacobian converts its components from the x coordinate system to the y coordinate system.

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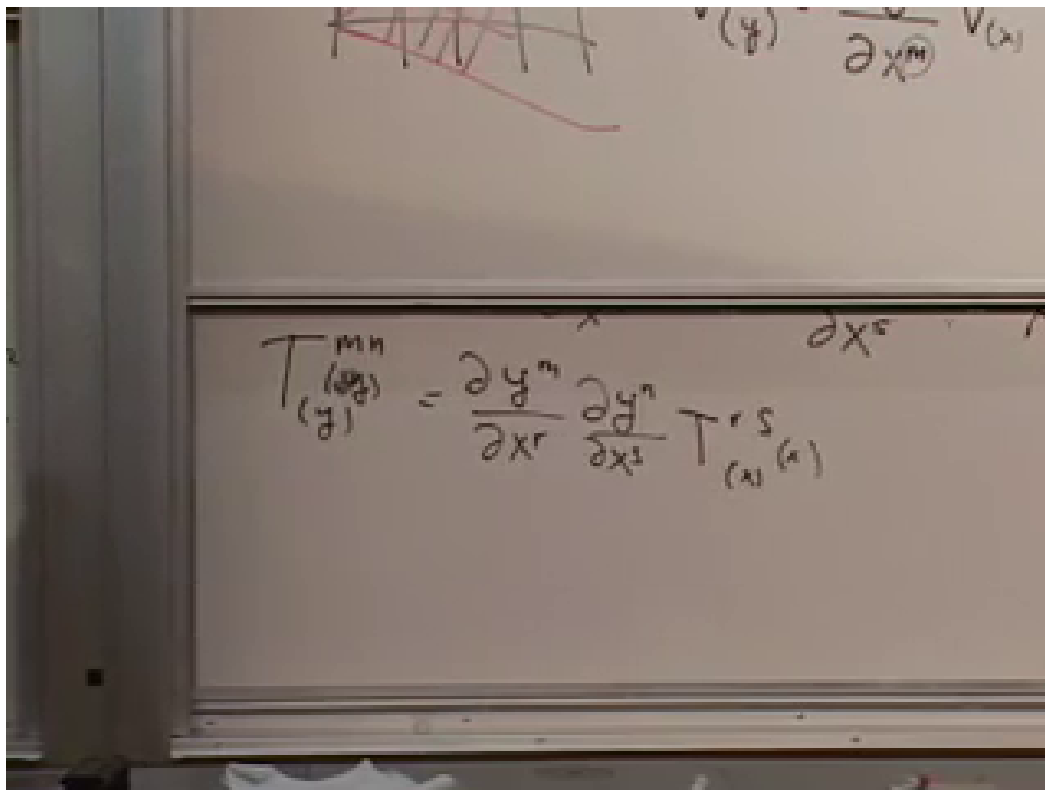


Figure 1.11: The rank-two contravariant transformation law. Each upper index contributes one factor of the direct Jacobian.

Lecture frame: 00:59:40.

Question from the audience. What is contravariant about a contravariant vector? ^a

Response. The name refers to its transformation behavior. Its components transform with the direct Jacobian $\partial y/\partial x$, whereas gradients and other covectors transform with the inverse Jacobian $\partial x/\partial y$. The distinction is encoded by upper and lower indices.

^aLecture timestamp: 00:51:25–00:52:36.

1.5 Higher-Rank Tensors and Covectors

If A^m and B^n are contravariant vectors, their component products provide a rank-two example,

$$T^{mn} = A^m B^n. \quad (1.15)$$

There are d^2 components, and each upper index brings its own direct Jacobian:

$$T_{(y)}^{pq} = \frac{\partial y^p}{\partial x^m} \frac{\partial y^q}{\partial x^n} T_{(x)}^{mn}. \quad (1.16)$$

Not every rank-two tensor need be a single product of two vectors; the product is a convenient way to discover the transformation pattern.

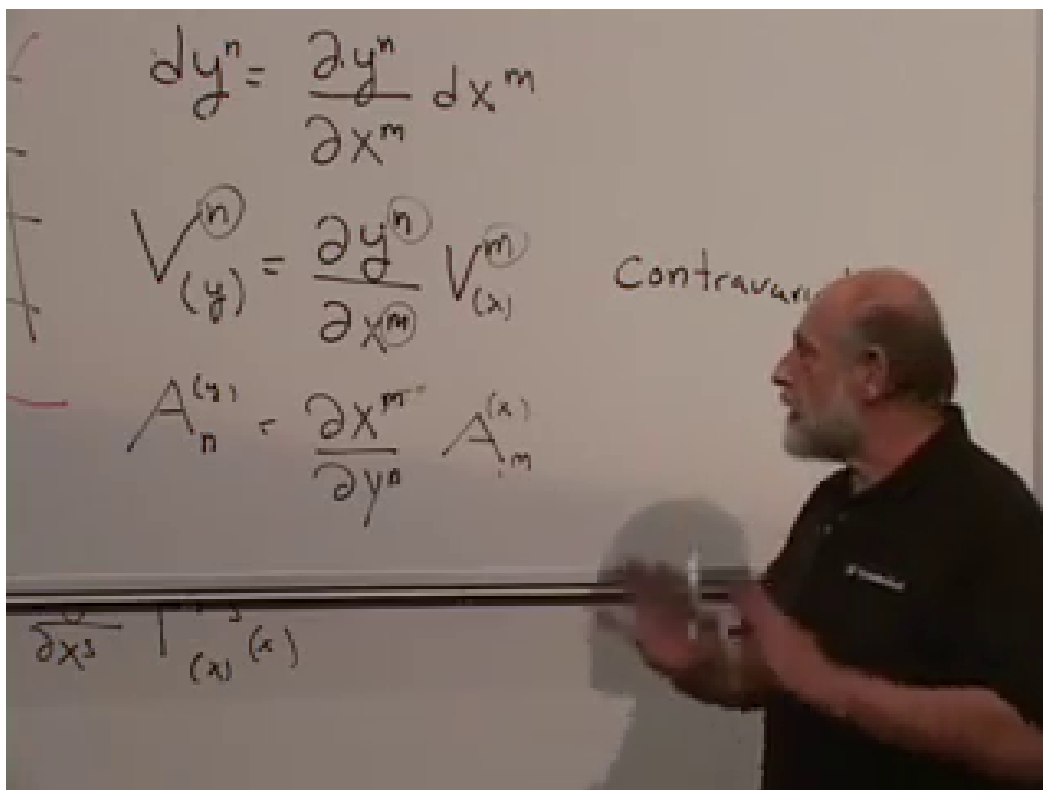


Figure 1.12: Contravariant and covariant vector laws displayed together. The direct and inverse Jacobians explain the upper- and lower-index notation.

Lecture frame: 01:07:40.

Question from the audience. What assumptions are hidden in this manipulation? ^a

Response. The components are ordinary numbers, so multiplication commutes. The coordinate functions are differentiable, and the Jacobian is invertible because the coordinate map is locally one-to-one. These are the same regularity assumptions already used in the chain rule.

^aLecture timestamp: 01:00:00–01:03:12.

The gradient of a scalar transforms oppositely. Define

$$A_m^{(x)} = \frac{\partial \phi}{\partial x^m}. \quad (1.17)$$

Equation (1.10) says

$$A_n^{(y)} = \frac{\partial x^m}{\partial y^n} A_m^{(x)}. \quad (1.18)$$

This is a covariant vector, or covector. Contravariant components carry upper indices; covariant components carry lower indices.

An upper index can be contracted with a lower index:

$$A_n V^n. \quad (1.19)$$

Under a coordinate transformation the two Jacobians cancel by Eq. (1.11), leaving a scalar. This is why the placement of indices is more than typography.

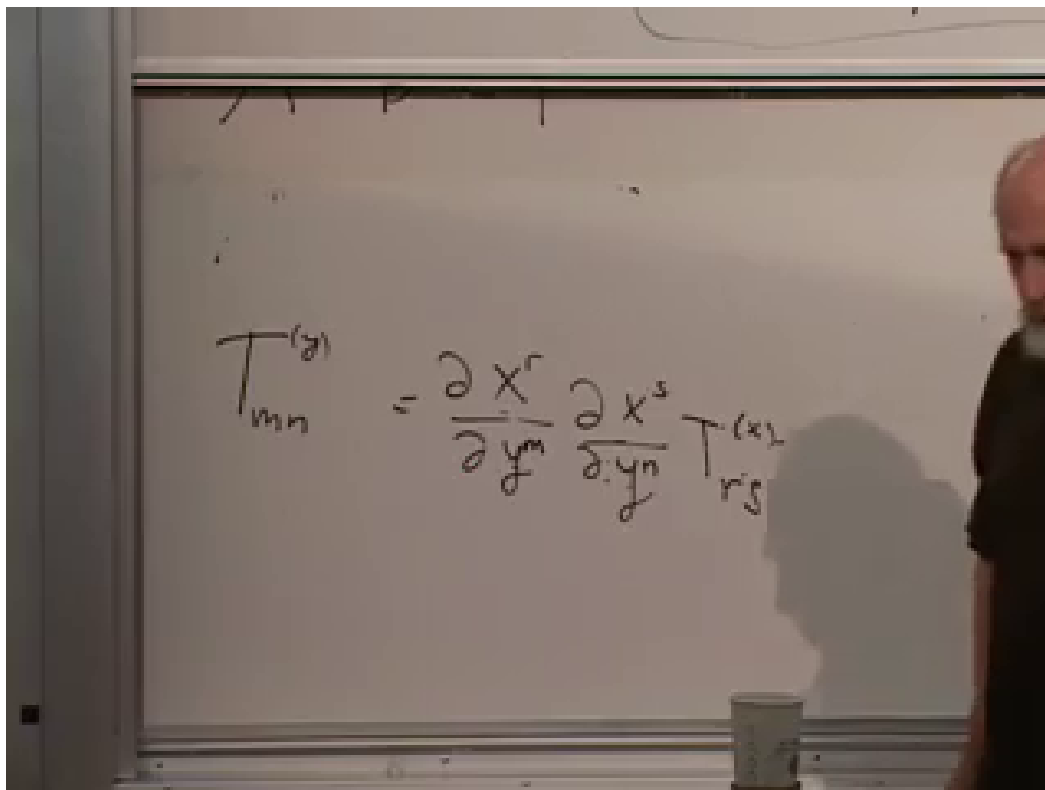


Figure 1.13: The rank-two covariant transformation law, with one inverse Jacobian for each lower index.

Lecture frame: 01:11:45.

With two lower indices the transformation rule is

$$T_{mn}^{(y)} = \frac{\partial x^r}{\partial y^m} \frac{\partial x^s}{\partial y^n} T_{rs}^{(x)}. \quad (1.20)$$

Question from the audience. Are upper and lower indices merely names for two different lists of numbers? ^a

Response. They label two dual transformation laws. A vector and a covector are different geometric kinds of object, although the metric will soon provide a systematic map between them. The index placement also tells us which contractions are coordinate invariant.

^aLecture timestamp: 01:09:15–01:11:42.

1.6 The Metric as a Tensor

After the break, begin again with the most familiar geometry: Euclidean space in Cartesian coordinates. For an infinitesimal displacement, Pythagoras gives

$$ds^2 = (dx^1)^2 + (dx^2)^2 + \cdots + (dx^d)^2. \quad (1.21)$$

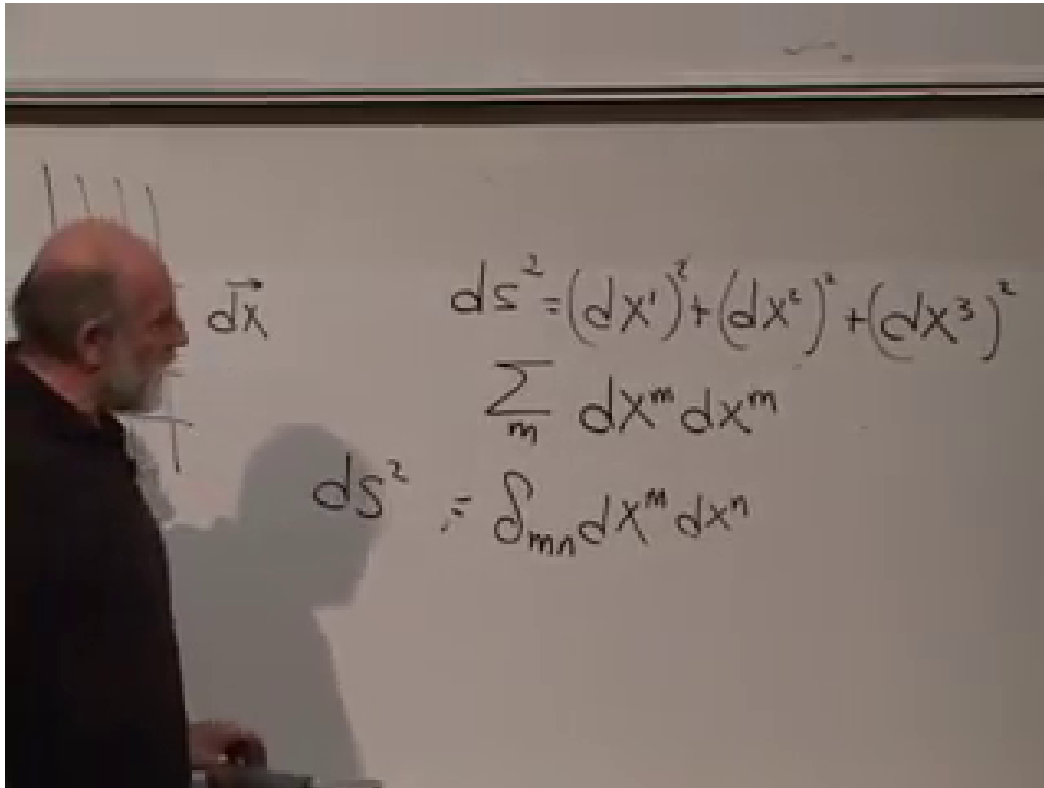


Figure 1.14: Pythagoras and the Kronecker-delta form of the Cartesian line element. The metric measures the squared length of an infinitesimal displacement.

Lecture frame: 01:22:20.

The superscript on x^m labels a component; the superscript on ds^2 is an actual square. Introduce the Kronecker delta,

$$\delta_{mn} = \begin{cases} 1, & m = n, \\ 0, & m \neq n, \end{cases} \quad (1.22)$$

and write

$$ds^2 = \delta_{mn} dx^m dx^n. \quad (1.23)$$

Now change from x to y . The differentials obey

$$dx^m = \frac{\partial x^m}{\partial y^r} dy^r. \quad (1.24)$$

Substitution into Eq. (1.23) gives

$$\begin{aligned} ds^2 &= \delta_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} dy^r dy^s \\ &= g_{rs}^{(y)}(y) dy^r dy^s, \end{aligned} \quad (1.25)$$

where

$$g_{rs}^{(y)}(y) = \delta_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s}. \quad (1.26)$$

The image shows a chalkboard with three handwritten equations. The first equation is $dx^m = \frac{\partial x^m}{\partial y^r} dy^r$. To its right is the label dy . The second equation is $ds^2 = \left(\sum_{mn} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s} \right) dy^s dy^r$, where the summation term is circled. The third equation is $g_{rs}(y) dy^s dy^r$, where the entire expression is circled.

Figure 1.15: The Cartesian line element transformed to arbitrary coordinates. The circled coefficient is the metric $g_{rs}^{(y)}$.

Lecture frame: 01:28:10.

More generally, if the metric in the x -coordinates is $g_{mn}^{(x)}$, then

$$g_{rs}^{(y)} = g_{mn}^{(x)} \frac{\partial x^m}{\partial y^r} \frac{\partial x^n}{\partial y^s}. \quad (1.27)$$

The metric is therefore a rank-two covariant tensor. The number ds^2 is invariant, while the components g_{mn} and dx^m change in compensating ways.

1.7 Flat Space, Curved Space, and Local Coordinates

A space is flat on a region if coordinates can be found throughout that region in which

$$g_{mn} = \delta_{mn}. \quad (1.28)$$

Complicated, position-dependent metric components do not by themselves prove curvature; they may arise from curvilinear coordinates on a flat space. Curvature is the obstruction to transforming the metric to the Cartesian form over the region under consideration.

Question from the audience. So is flatness exactly the existence of coordinates that reduce the metric to δ_{mn} ? ^a

Response. Yes, provided the region is specified. Flatness means that one coordinate transformation works throughout that region. A metric that merely looks complicated in the current chart may still describe flat space.

^aLecture timestamp: 01:31:20–01:34:22.

At any regular point of a smooth Riemannian geometry, coordinates can be chosen so that the metric equals δ_{mn} at that point, and even so that its first derivatives vanish there. This is a pointwise statement, not a claim that a finite neighborhood is exactly Euclidean. Curvature appears in the second-order departure from that locally Cartesian form.

The cone is a useful warning about local and global statements. Away from its tip, a small patch can be cut and unrolled into the plane, so its intrinsic geometry is locally flat. At the tip there is a conical singularity, and globally the missing or excess angle cannot be removed by one smooth Cartesian chart.

Question from the audience. Can the metric always be made Euclidean in a sufficiently small neighborhood? ^a

Response. It can always be made Euclidean at one regular point. Over a finite neighborhood, exact Euclidean form is possible only when the curvature vanishes there. The cone is locally flat away from its tip, but a genuinely curved surface such as a sphere retains second-order curvature in every neighborhood.

^aLecture timestamp: 01:34:22–01:37:18.

1.8 Polar Coordinates in the Flat Plane

Polar coordinates supply the decisive example of a nonconstant metric in a flat space:

$$x = r \cos \theta, \quad y = r \sin \theta. \quad (1.29)$$

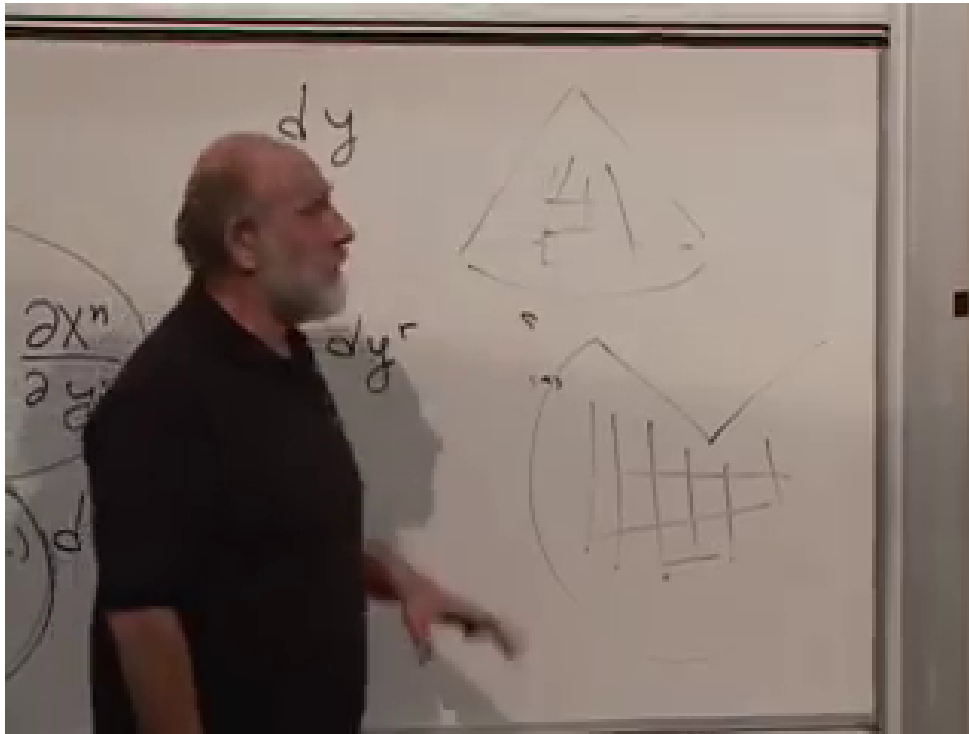


Figure 1.16: A cone and a locally flattened patch. The construction separates pointwise or local flatness from a global obstruction concentrated at the cone's tip.

Lecture frame: 01:36:40.

Differentiate:

$$\begin{aligned} dx &= \cos \theta dr - r \sin \theta d\theta, \\ dy &= \sin \theta dr + r \cos \theta d\theta. \end{aligned} \tag{1.30}$$

Substituting and expanding,

$$\begin{aligned} ds^2 &= dx^2 + dy^2 \\ &= (\cos \theta dr - r \sin \theta d\theta)^2 \\ &\quad + (\sin \theta dr + r \cos \theta d\theta)^2 \\ &= dr^2 + r^2 d\theta^2. \end{aligned} \tag{1.31}$$

The mixed terms cancel. Thus

$$(g_{ab}) = \begin{pmatrix} g_{rr} & g_{r\theta} \\ g_{\theta r} & g_{\theta\theta} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}. \tag{1.32}$$

Question from the audience. Does the vanishing off-diagonal term prove that the plane is flat? ^a

Response. No. It says that the r - and θ -coordinate curves meet orthogonally. Diagonal form is useful, but curvature is a different question. The plane is flat because the polar metric can be transformed back to $dx^2 + dy^2$.

^aLecture timestamp: 01:45:15–01:47:05.

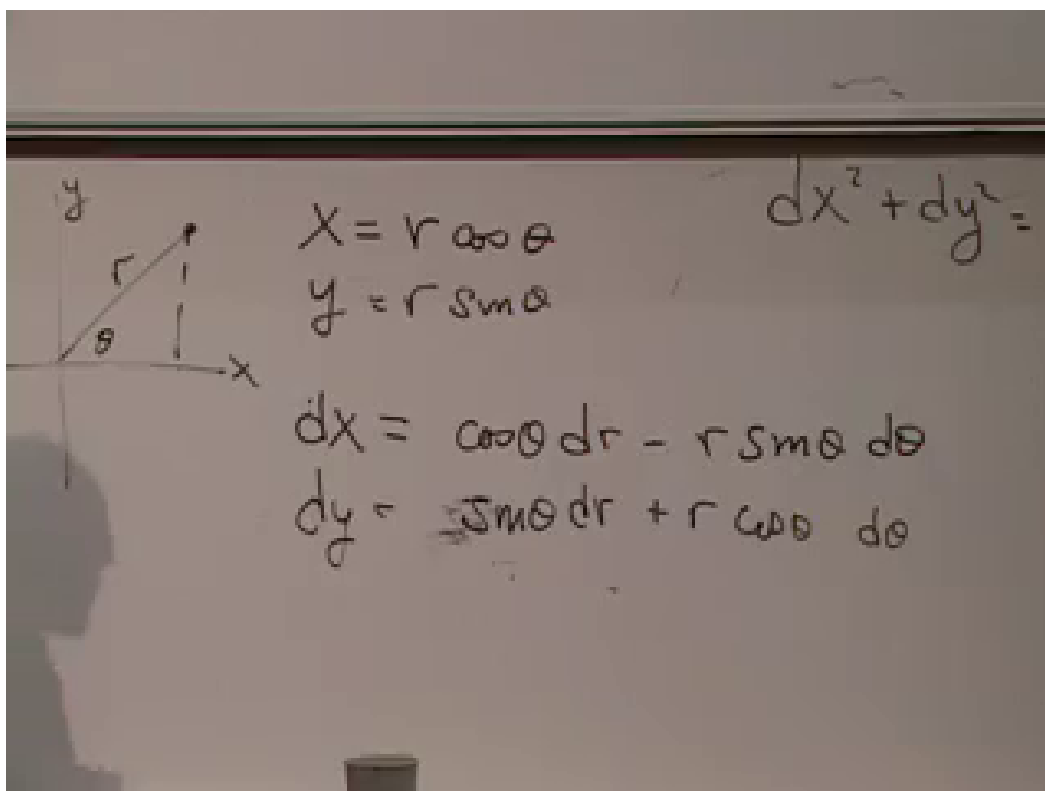


Figure 1.17: The Cartesian-to-polar map and its differential form. These relations are substituted directly into $dx^2 + dy^2$.

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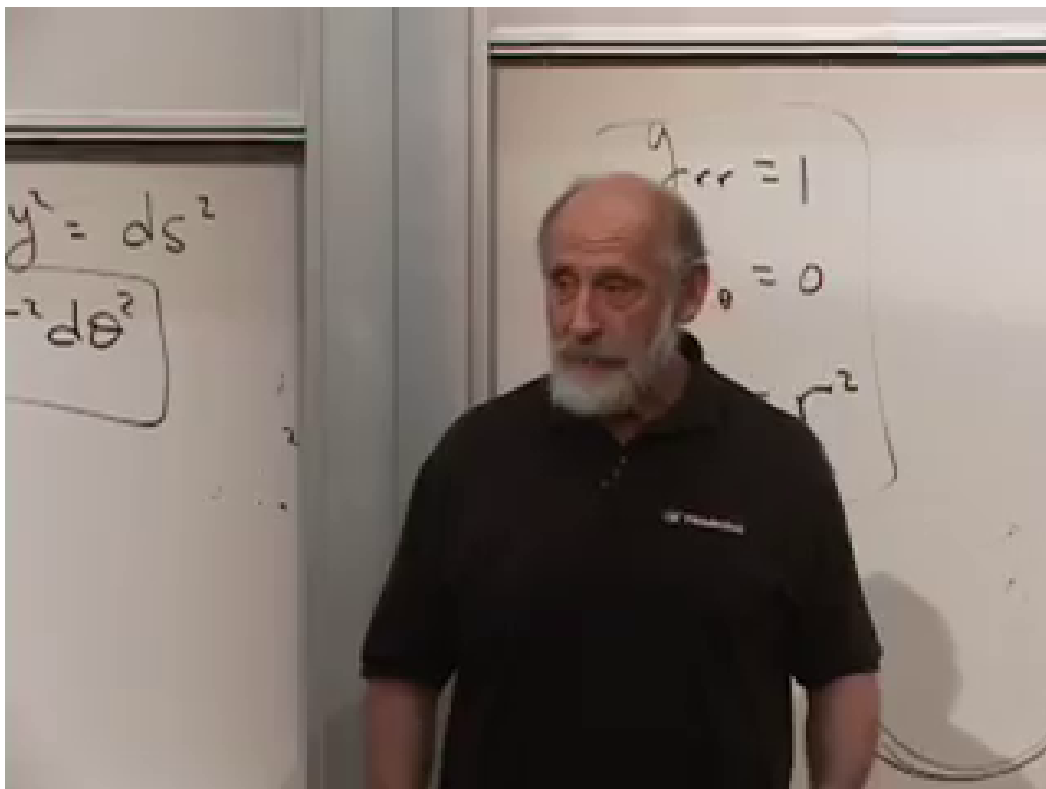


Figure 1.18: The completed polar-coordinate metric: $g_{rr} = 1$, $g_{r\theta} = 0$, and $g_{\theta\theta} = r^2$.
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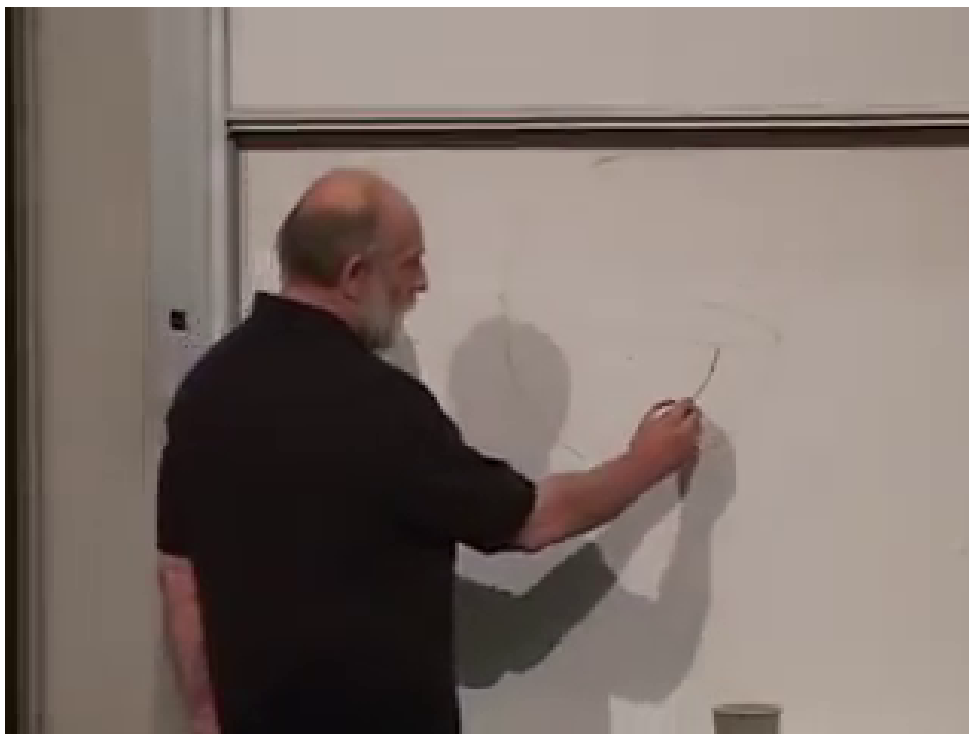


Figure 1.19: Polar and azimuthal displacements on a sphere. The shrinking latitude-circle radius supplies the factor $\sin \phi$ in the metric.

Lecture frame: 01:49:30.

1.9 The Metric on a Sphere

Finally take a unit sphere. Let ϕ be the polar angle measured from the north pole and let θ be the azimuthal angle. A displacement in the ϕ direction has length $d\phi$. At fixed ϕ , the circle of latitude has radius $\sin \phi$, so an azimuthal displacement has length $\sin \phi d\theta$. The two coordinate directions are orthogonal, and

$$ds^2 = d\phi^2 + \sin^2 \phi d\theta^2. \quad (1.33)$$

For a sphere of radius R , the right-hand side is multiplied by R^2 .

The polar metric of the plane and the angular metric of the sphere now look similar, but they describe different intrinsic geometry. The plane's position-dependent coefficient can be removed by returning to Cartesian coordinates. The sphere's curvature cannot be removed over any finite patch. This is the distinction needed for gravity: coordinates may create apparent fields, but tidal effects reveal an invariant geometric obstruction. The metric tensor is the object that carries that geometry from one coordinate system to another.